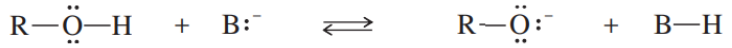
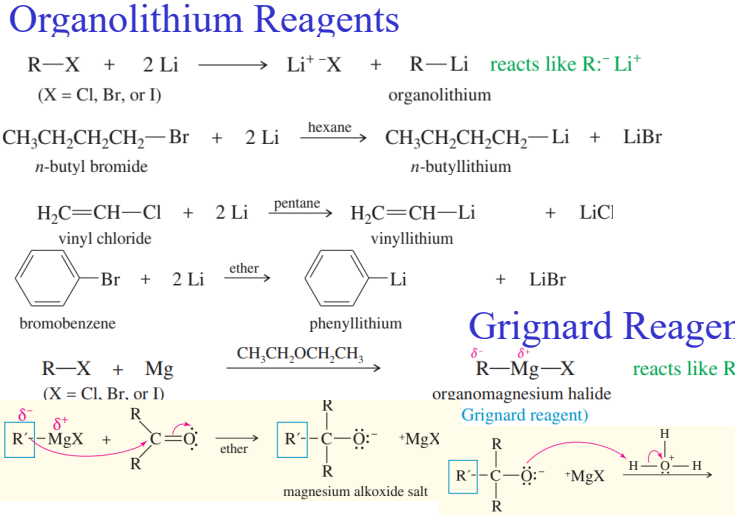
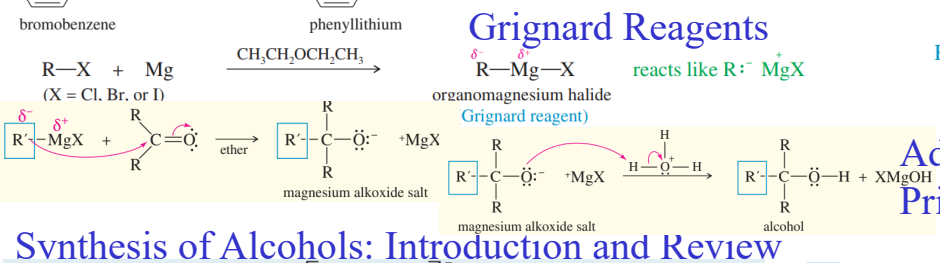
Acidity of Alcohols and Phenols



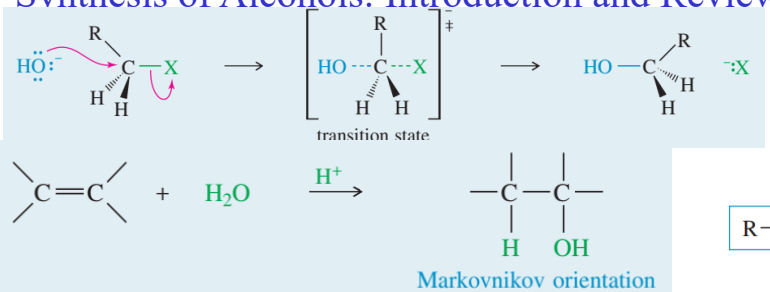
Organolithium Reagents

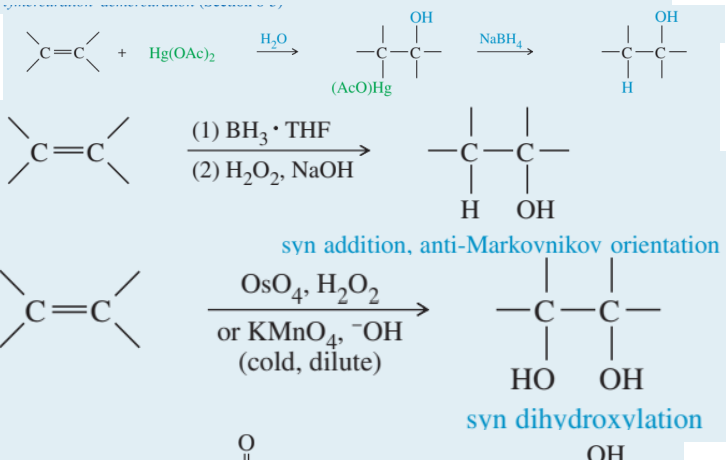


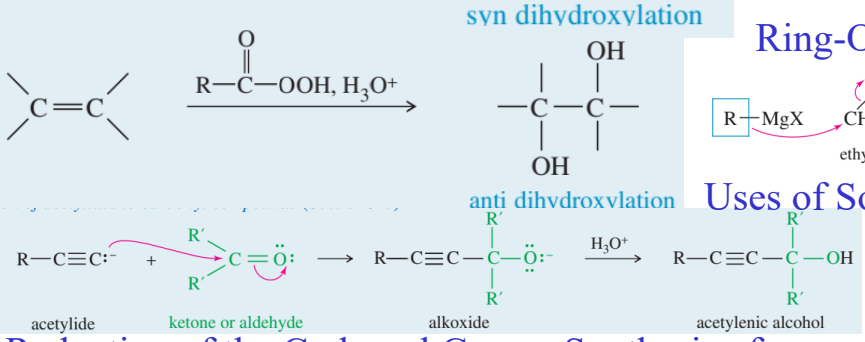
Grignard Reagents



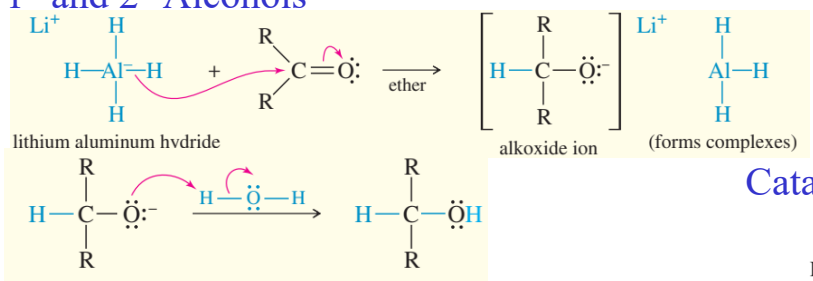
Synthesis of Alcohols: Introduction and Review



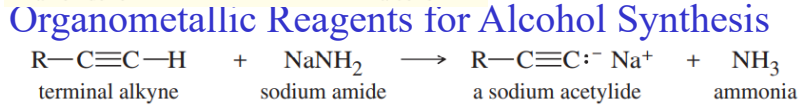




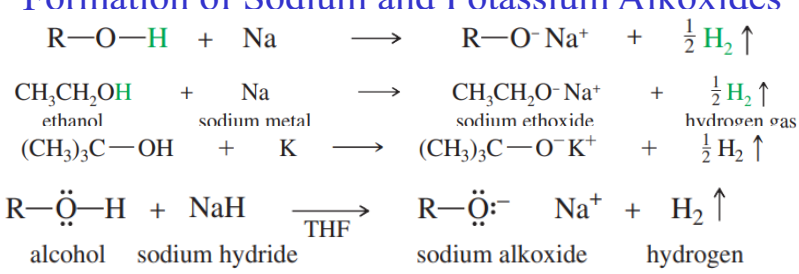
Reduction of the Carbonyl Group: Synthesis of 1° and 2° Alcohols



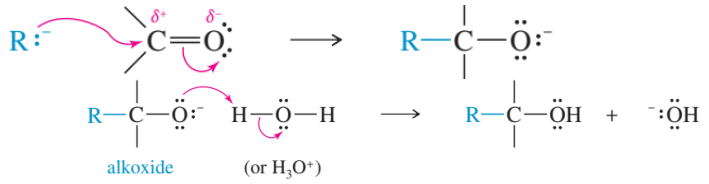
Organometallic Reagents for Alcohol Synthesis



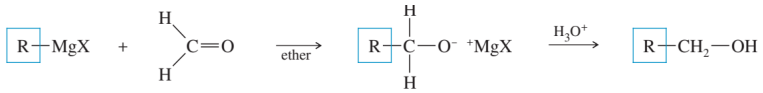
Formation of Sodium and Potassium Alkoxides



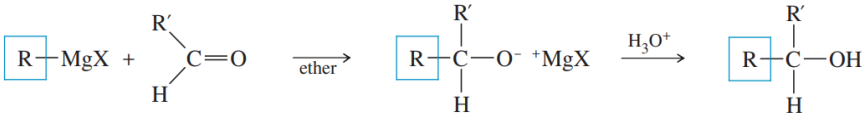
Reactions of Organometallic Compounds



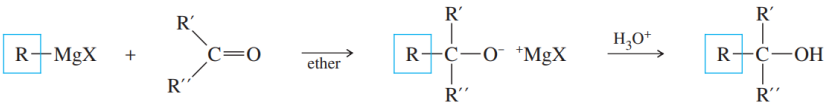
Addition to Formaldehyde: Formation of Primary Alcohols



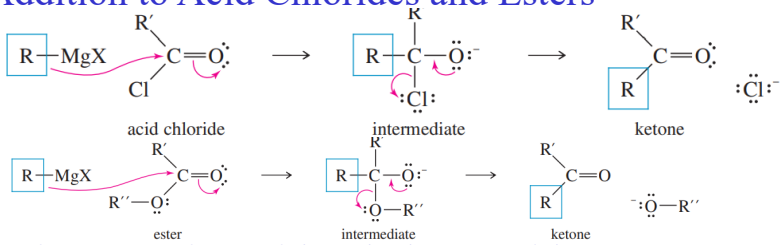
Addition to Aldehydes: Formation of Secondary Alcohols



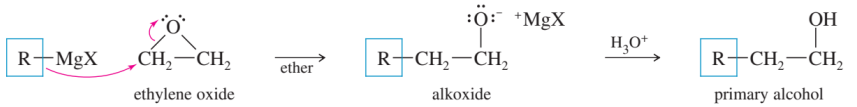
Addition to Ketones: Formation of Tertiary Alcohols



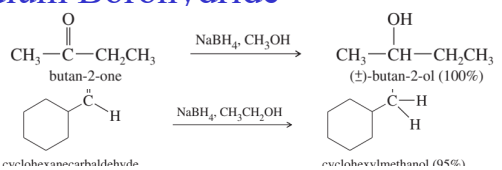
Addition to Acid Chlorides and Esters

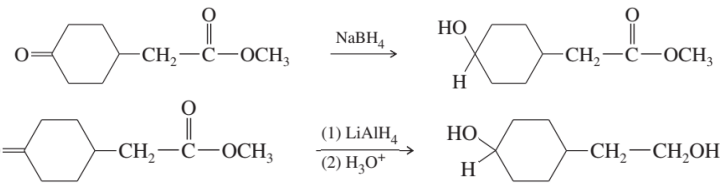


Ring-Opening with Ethylene Oxide

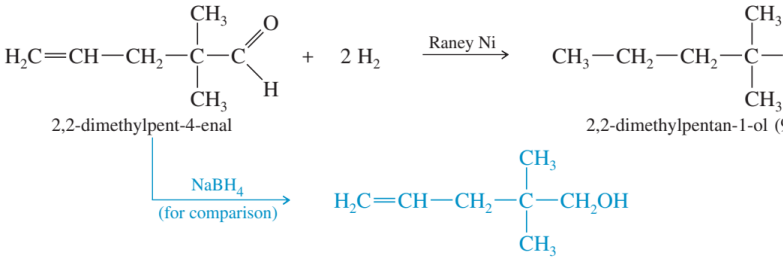


Uses of Sodium Borohydride

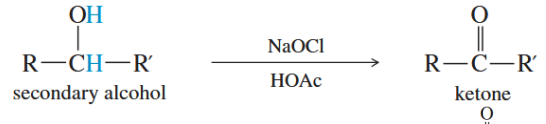


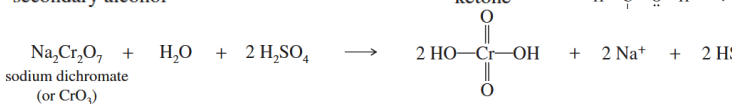


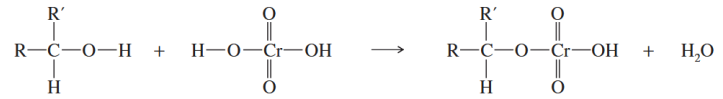
Catalytic Hydrogenation of Ketones and Aldehydes

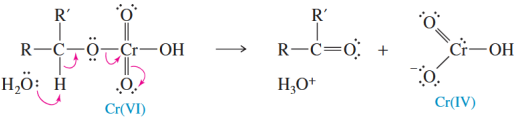


Oxidation of Secondary Alcohols

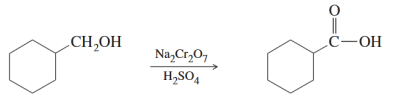


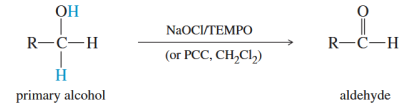


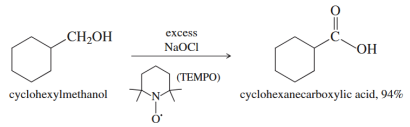


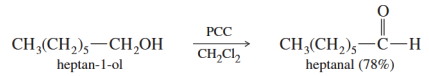


Oxidation of Primary Alcohols

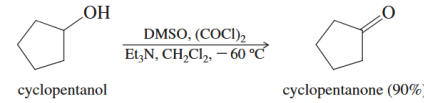


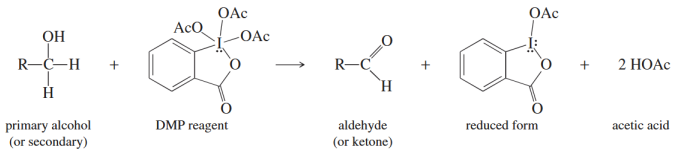






Additional Methods for Oxidizing Alcohols

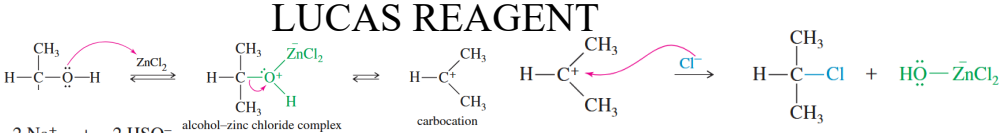




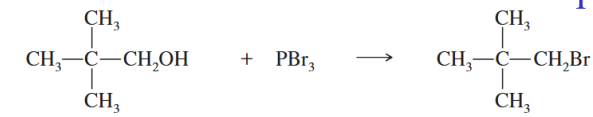
Reactions with Hydrochloric Acid

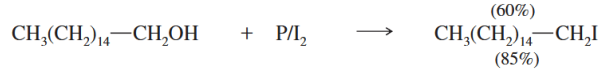
$$R-OH + HCl/H_2O \xrightarrow{ZnCl_2} R-Cl$$

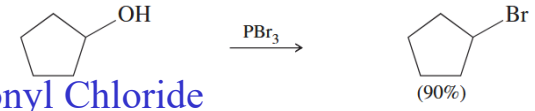
LUCAS REAGENT



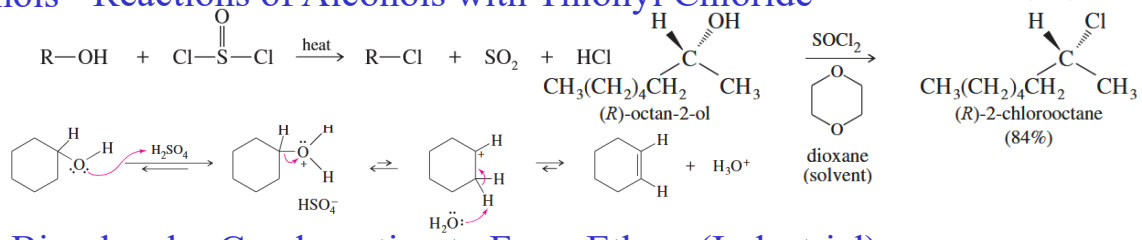
Reactions of Alcohols with Phosphorus Halides



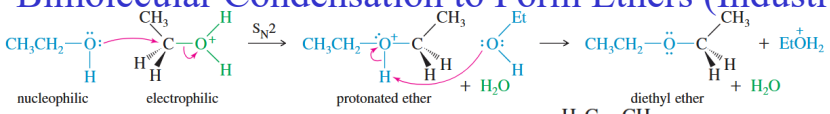




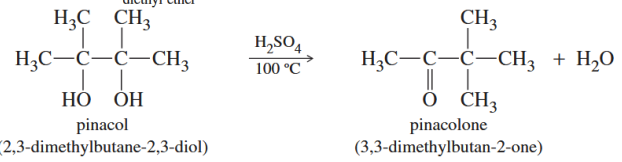
Reactions of Alcohols with Thionyl Chloride



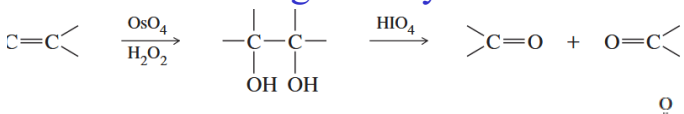
Bimolecular Condensation to Form Ethers (Industrial)



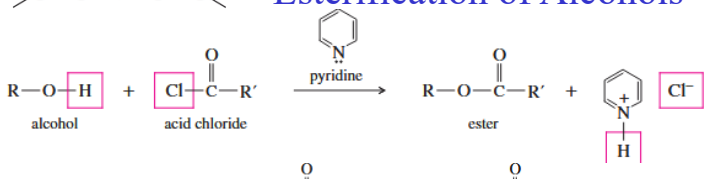
The Pinacol Rearrangement



Periodic Acid Cleavage of Glycols



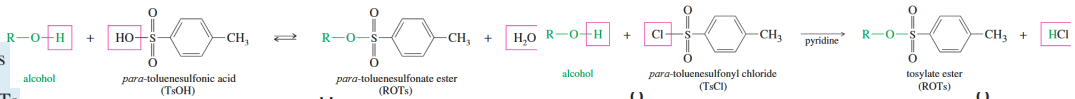
Esterification of Alcohols



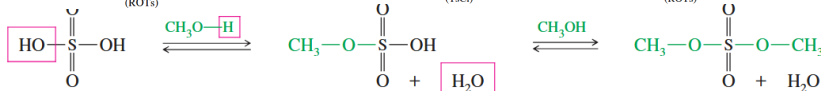
Alcohols as Nucleophiles and Electrophiles; Formation of Tosylates

R—OTs	+	⁻OH (hydroxide)	⟶	R—OH (alcohol)	+	⁻OTs
R—OTs	+	⁻C≡N (cyanide)	⟶	R—C≡N (nitrile)	+	⁻OTs
R—OTs	+	Br^- (halide)	⟶	R—Br (alkyl halide)	+	⁻OTs
R—OTs	+	R′—O⁻ (alkoxide)	⟶	R—O—R′ (ether)	+	⁻OTs
R—OTs	+	:NH_3 (ammonia)	⟶	R—NH_3^+ (amine salt)		⁻OTs
R—OTs	+	$LiAlH_4$ (LAH)	⟶	R—H (alkane)	+	⁻OTs

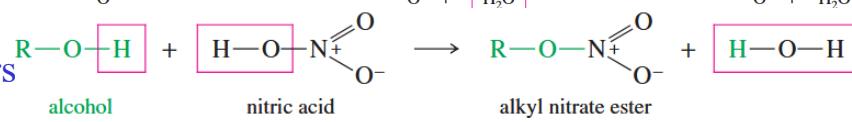
Esters of Inorganic Acids



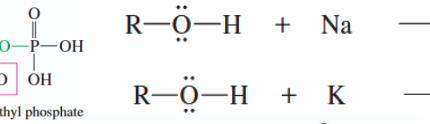
Sulfate Esters

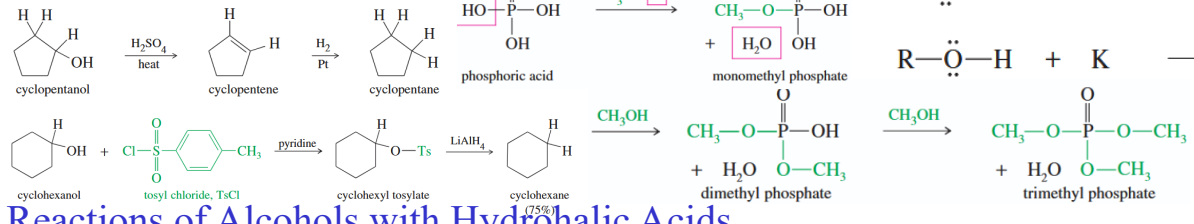


Nitrate Esters



Phosphate Esthers



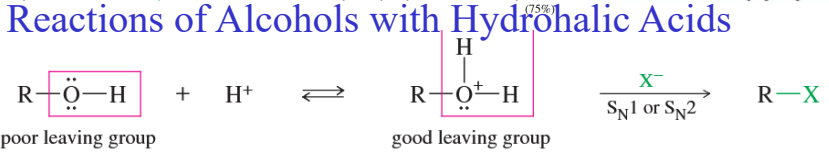


Reactions of Alkoxides

$$R-\ddot{\underset{..}{O}}-H + Na \longrightarrow R-\ddot{\underset{..}{O}}:^- \; Na^+ + \tfrac{1}{2} H_2 \uparrow$$

$$R-\ddot{\underset{..}{O}}-H + K \longrightarrow R-\ddot{\underset{..}{O}}:^- \; K^+ + \tfrac{1}{2} H_2 \uparrow$$

Reactions of Alcohols with Hydrohalic Acids



Reactions with Hydrobromic Acid

$$R-OH + HBr/H_2O \longrightarrow R-Br$$

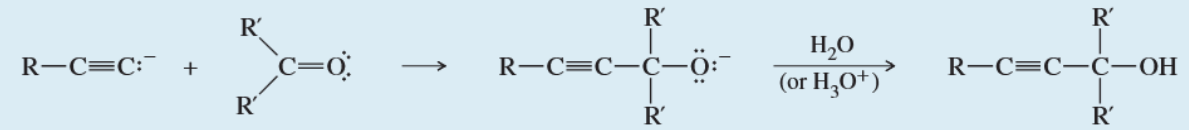
Chapter nine

ACETYLIDE CHEMISTRY

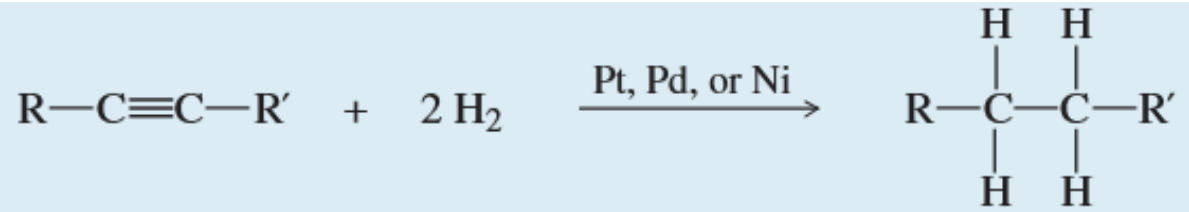
$$R{-}C{\equiv}C{-}H + NaNH_2 \longrightarrow R{-}C{\equiv}C{:}^{-}\,Na^{+} + NH_3$$
$$R{-}C{\equiv}C{-}H + R'Li \longrightarrow R{-}C{\equiv}CLi + R'H$$
$$R{-}C{\equiv}C{-}H + R'MgX \longrightarrow R{-}C{\equiv}CMgX + R'{-}H$$

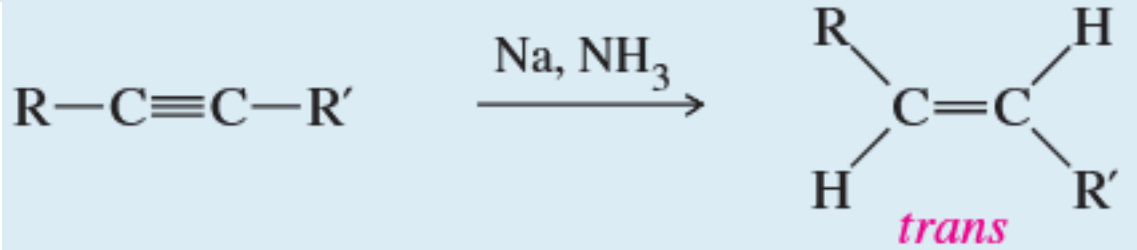
$$R{-}C{\equiv}C{:}^{-} + R'{-}X \longrightarrow R{-}C{\equiv}C{-}R'$$

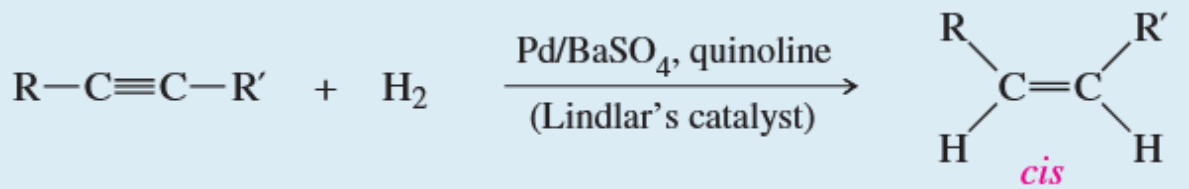
(R′—X must be an unhindered primary halide or tosylate.)

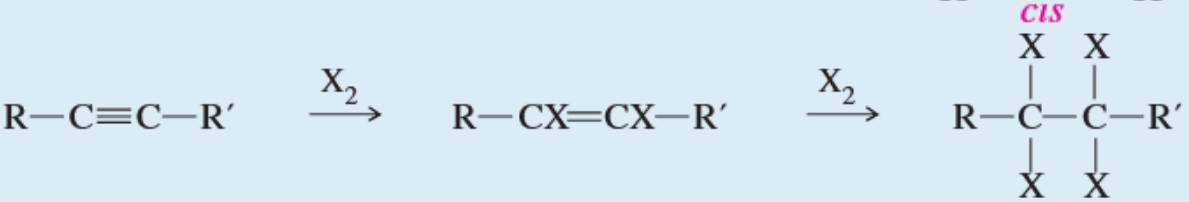


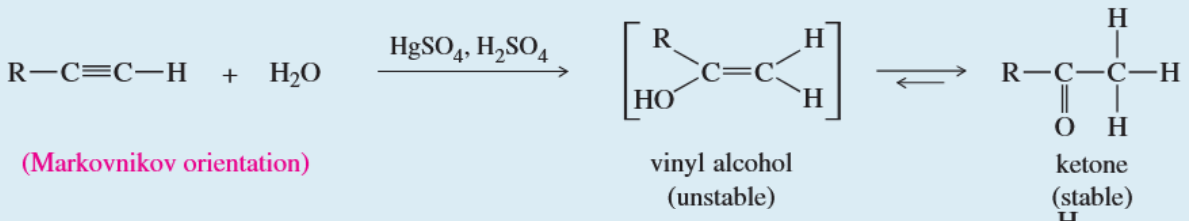
ADDITIONS TO THE TRIPLE BOND

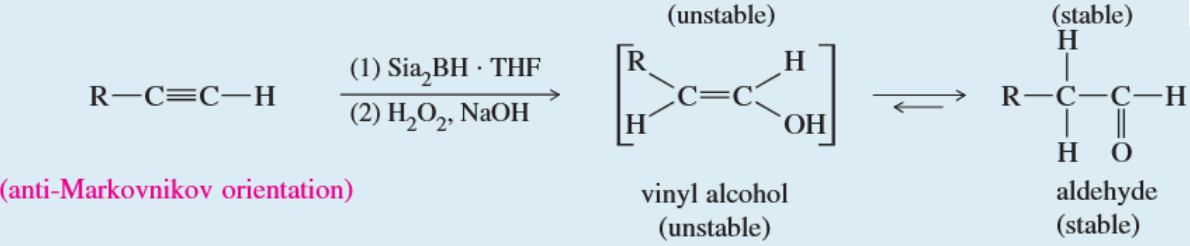




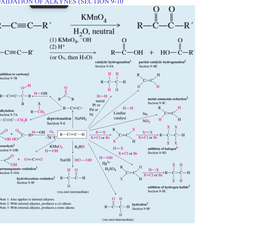








OXIDATION OF ALKYNES (SECTION 9-10)

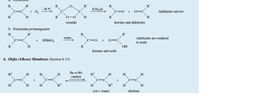


Note 1: Also applies to internal alkynes.
Note 2: With internal alkynes, produces a cis alkene.
Note 3: With internal alkynes, produces a trans alkene.

Chapter 8 summary alkenes

1. *Electrophilic Additions*

a. *Addition of hydrogen halides* (Section 8-3)



works with HCl, HBr, and HI
Markovnikov orientation
(anti-Markovnikov
with HBr and peroxides)
Rearrangements are possible

(HX = HCl, HBr, or HI)

b. *Acid-catalyzed hydration* (Section 8-4)

Markovnikov orientation
Rearrangements are possible

c. *Oxymercuration-demercuration* (Section 8-5)

Markovnikov orientation
no rearrangements

d. *Alkoxymercuration-demercuration* (Section 8-6)

Markovnikov orientation

e. *Hydroboration-oxidation* (Section 8-7)

anti Markovnikov orientation
(syn addition)
no rearrangements

f. *Polymerization* (Section 8-16)

cationic, radical, or
anionic polymerization

2. *Reduction Catalytic Hydrogenation* (Section 8-10)

syn addition

3. *Addition of Carbenes: Cyclopropanation* (Section 8-11)

a. *Simmons-Smith Reaction*

(or enantiomer)

cyclohexene — 3-hexenes (cis/trans)

Active concentration of ZnI_2 in diethyl substantially reduces the rate of the reaction. Rate is proportional to the concentration of the active species, which is the rate-limiting step. (Section 8-11B)

b. *Dichlorocarbene Formation* (Section 8-9)

anti addition
Markovnikov orientation
Br^- adds the C^+ would

c. *Dihalohalocarbene* (Section 8-12)

alkene — gem-trihalide — syn addition

d. *Anti alkylmetalation* (Section 8-13)

anti addition

e. *Syn alkylmetalation* (Section 8-14)

syn addition

3. *Oxidative Cleavage of Alkenes* (Section 8-15)

a. *Ozonolysis*

ozonide — ketones and aldehydes — Aldehydes survive

b. *Permanent permanganate*

ketones and acids — Aldehydes are oxidized to acids

4. *Diels (Alkene) Blomhan* (Section 8-17)

syn + trans — alkadiene